Regression Refresher

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Regression

- Regression is defined as a statistical method that aims to examine the **relations** between two or more variables of interest.
- Normally, we use regression to conduct **prediction**.
- In this case, we already have some **pre-existing beliefs** about the nature of the relations (e.g., what comes first and what comes after and, sometimes, direction (i.e., positive or negative)), which makes regression different from correlation.

Regression

- What is the relation between smoking and health?
    - Does smoking predict health?
    - Smoking comes first and health comes after.
    - negative (smoke more > worse health)
- What is the relation between income and happiness ?
    - Does income predict happiness?
    - Income comes first and happiness comes after.
    - positive (more income > more happiness)

Activity

- What are some important and interesting relations you would like to explore in your area?
- What is the nature of that relation?

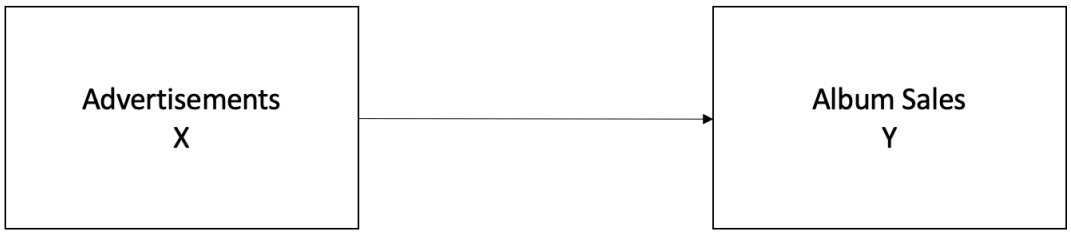
Notation

- What is the relation between advertisements and sales?
    - Do advertisements predict sales?
    - Advertisements come first and sales come after.
    - positive

Notation

- Rectangular boxes: manifest variables
- Arrows: temporal order between variables
- Rectangular boxes on the left side: predictors or independent variables or **X** (**what comes first**)

- Rectangular boxes on the right side: outcome variables or dependent variables or **Y** (**what comes after**)



Activity

- Use the notation shown here to draw the relation you would like to investigate.

Formula for Simple Linear Regression Coefficient

The formula for the slope coefficient β_1 in simple linear regression is:

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

The intercept is given by:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Standard Error of the Slope Coefficient

The standard error of the slope β_1 is:

$$SE_{\beta_1} = \frac{s}{\sqrt{\sum(x_i - \bar{x})^2}}$$

where

$$s = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}}$$

is the standard error of the regression.

Test Statistic

$$t = \frac{\beta_1}{SE_{\beta_1}}$$

Hypothesis Testing:

Null Hypothesis $H_0: \beta = 0$

(There is no linear relationship between the two variables in the population.)

Alternative Hypothesis $H_1: \beta \neq 0$

(There is a linear relationship between the two variables in the population.)

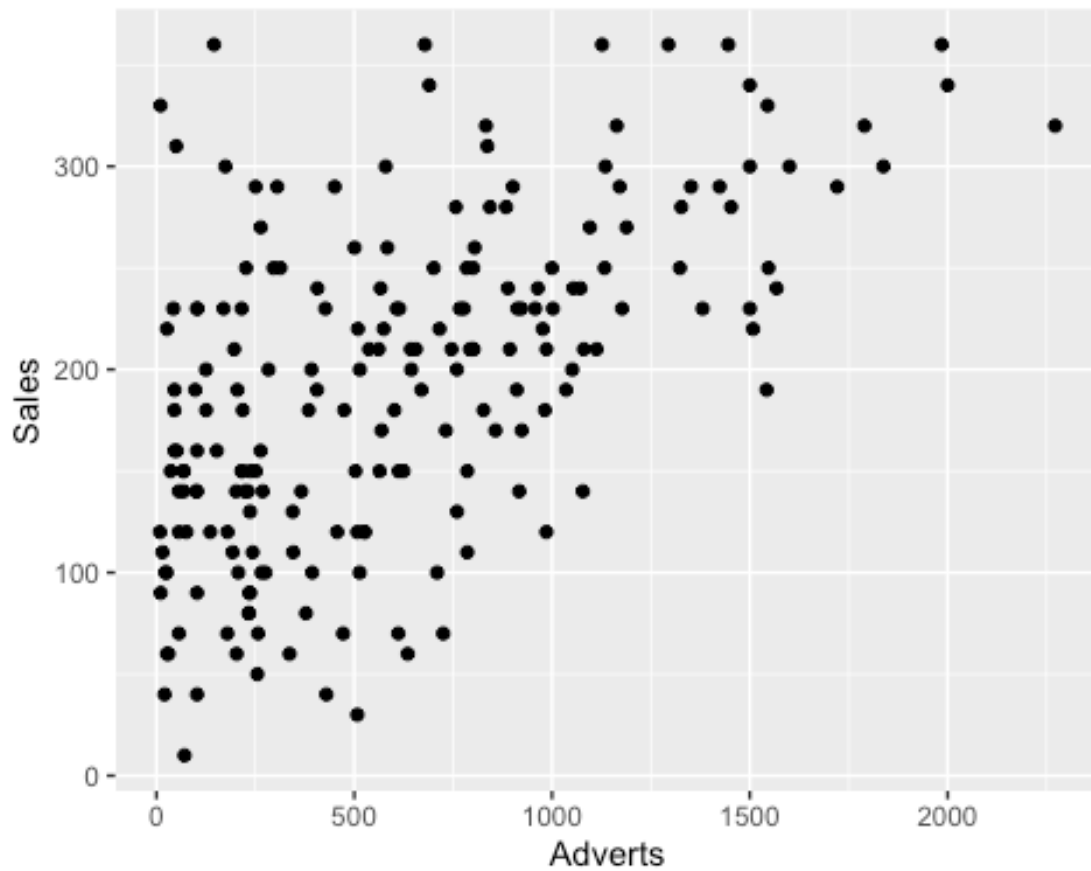
Load Packages and Import Data

```
library(tidyverse)
library(psych)
library(haven)
data <- read_sav("Album Sales.sav")

#view our data
data

# A tibble: 200 × 4
   Adverts Sales Airplay Image
     <dbl> <dbl>   <dbl> <dbl>
 1    10.3   330      43    10
 2   986.    120      28     7
 3  1446.    360      35     7
 4  1188.    270      33     7
 5   575.    220      44     5
 6   569.    170      19     5
 7   472.     70      20     1
 8   537.    210      22     9
 9   514.    200      21     7
10   174.    300      40     7
# ℹ 190 more rows

#descriptive statistics
describe(data)

        vars   n   mean     sd median trimmed    mad  min     max   range
skew
Adverts    1 200 614.41 485.66 531.92  560.81 489.09  9.1 2271.86 2262.76
0.84
Sales      2 200 193.20  80.70 200.00  192.69  88.96 10.0  360.00  350.00
0.04
Airplay    3 200  27.50  12.27  28.00   27.46  11.86  0.0   63.00   63.00
0.06
Image      4 200   6.77   1.40   7.00    6.88   1.48  1.0   10.00    9.00 -
1.27
        kurtosis    se
Adverts     0.17 34.34
Sales      -0.72  5.71
```

```
Airplay     -0.09  0.87
Image        3.56  0.10
```

Visualize

```
data |>
  ggplot(mapping = aes(x = Adverts, y = Sales)) +
  geom_point()
```



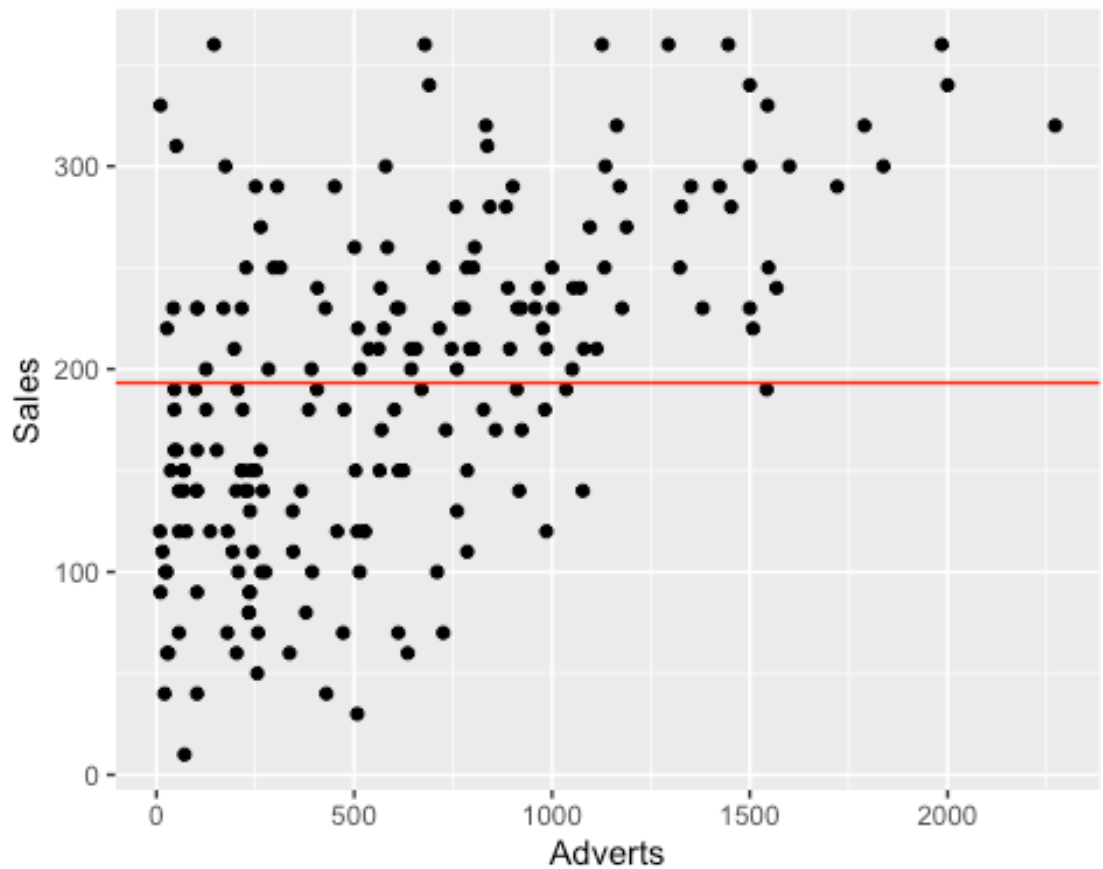
Model (simple linear regression: only one predictor)

```
fit<-lm(Sales ~ Adverts, data = data)
summary(fit)


Call:
lm(formula = Sales ~ Adverts, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-152.949  -43.796   -0.393   37.040  211.866 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
```

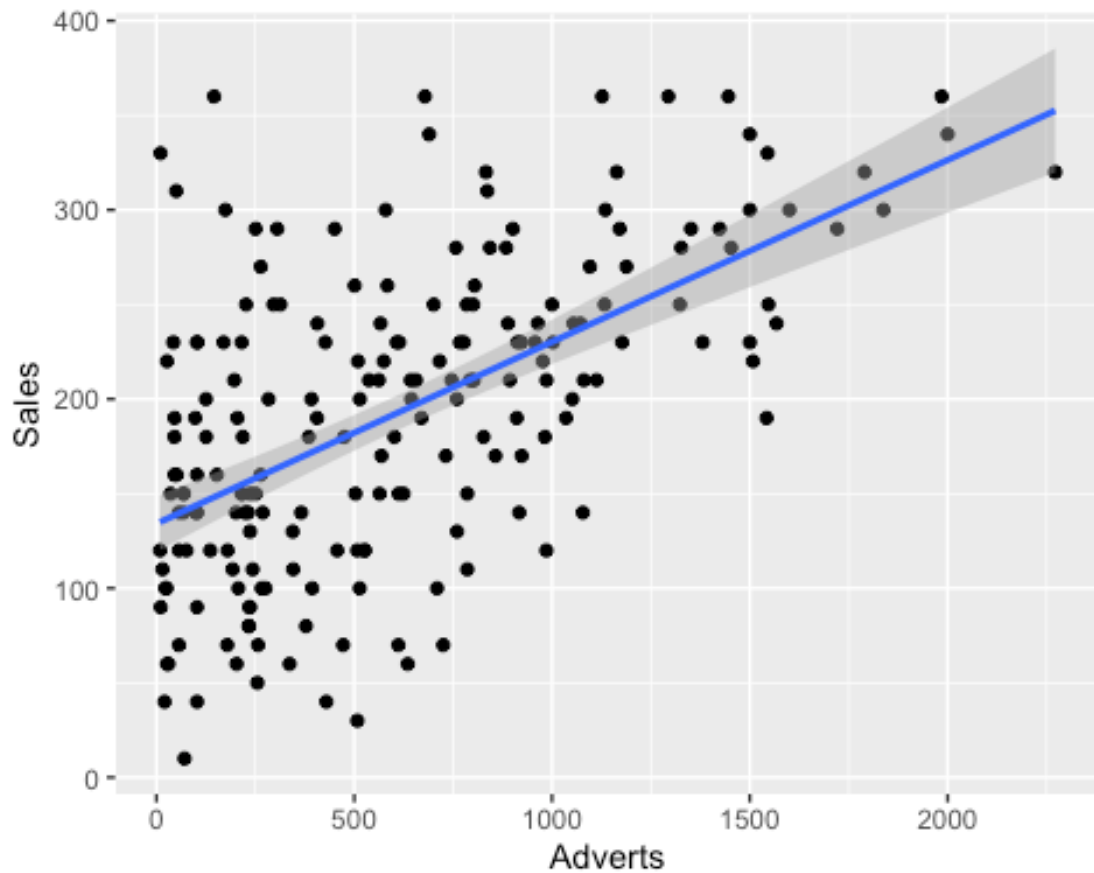
```
(Intercept) 1.341e+02  7.537e+00  17.799   <2e-16 ***
Adverts     9.612e-02  9.632e-03   9.979   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 65.99 on 198 degrees of freedom
Multiple R-squared:  0.3346,    Adjusted R-squared:  0.3313
F-statistic: 99.59 on 1 and 198 DF,  p-value: < 2.2e-16
```

```
#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 33.46% of variance in album sales is explained by
adverts.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Adverts significantly and positively predicts album sales.
One unit increase in adverts is expected to be associated with an increase of
0.09612 units in album sales.
```

```r
data |>
  ggplot(mapping = aes(x = Adverts, y = Sales)) +
  geom_point() +
  geom_hline(yintercept = 193.2, color = "red")
```




```
data |>
  ggplot(mapping = aes(x = Adverts, y = Sales)) +
  geom_point() +
  stat_smooth(method = "lm",
              formula = y ~ x,
              geom = "smooth")
```




```
#intercept = when predictor(s) are zero, what would y be? In this case, if we
don't invest any money in adverts, our sales are expected to be 134.10.
```

Regression Equation Example

$$Y_j = i + \beta 1 X_j + \varepsilon_j$$

$$Sales_j = 134.10 + 0.09612 Adverts_j + \varepsilon_j$$

```
#Suppose we invest $1000 in adverts, what would be the predicted album sales?

predict(fit, data.frame(Adverts = 1000))

       1
230.2644
```

Activity

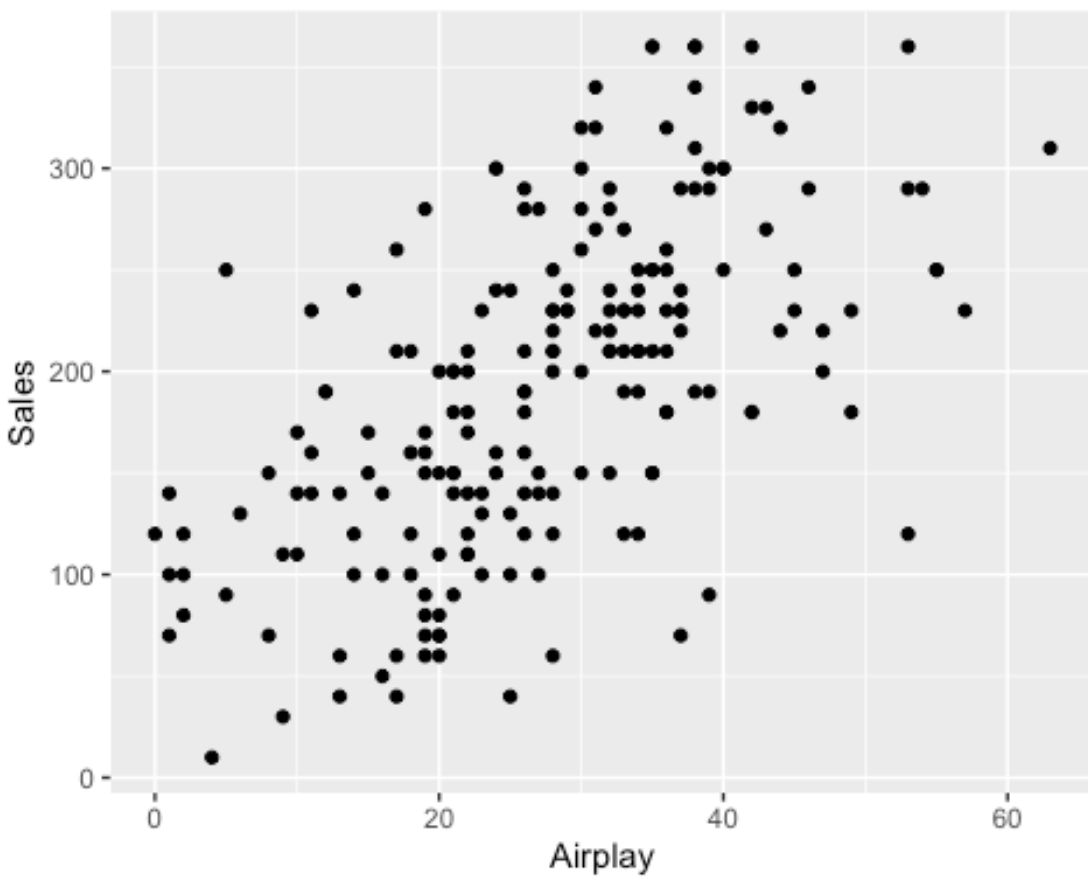
Let's try to use "Airplay" to predict album sales.

- Draw the diagram to represent the relation
- Visualize the relation between Airplay and Sales. Is it positive or negative?

- Run a simple linear regression model
- Evaluate the overall model, individual predictor, and intercept.
- Write down your interpretation of the results.
- If an album is played on the radio for 40 times, what would be the predicted album sales?

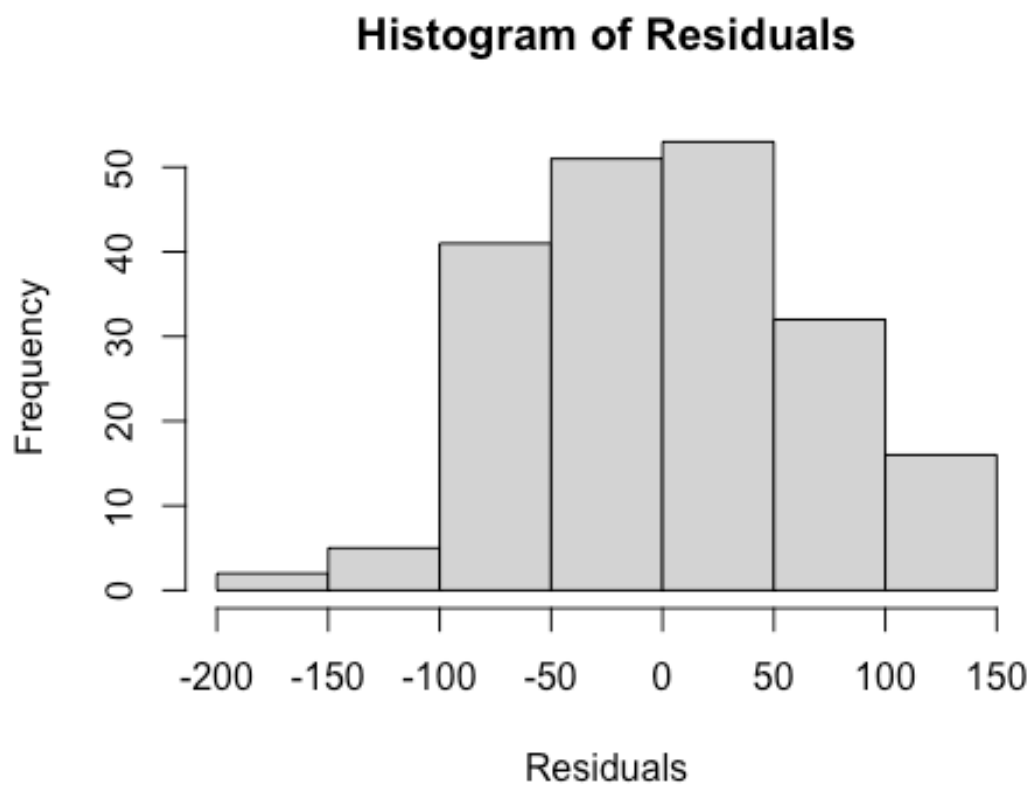
Solution

```
#visualize the data
data |>
  ggplot(mapping = aes(x = Airplay, y = Sales)) +
  geom_point()
```

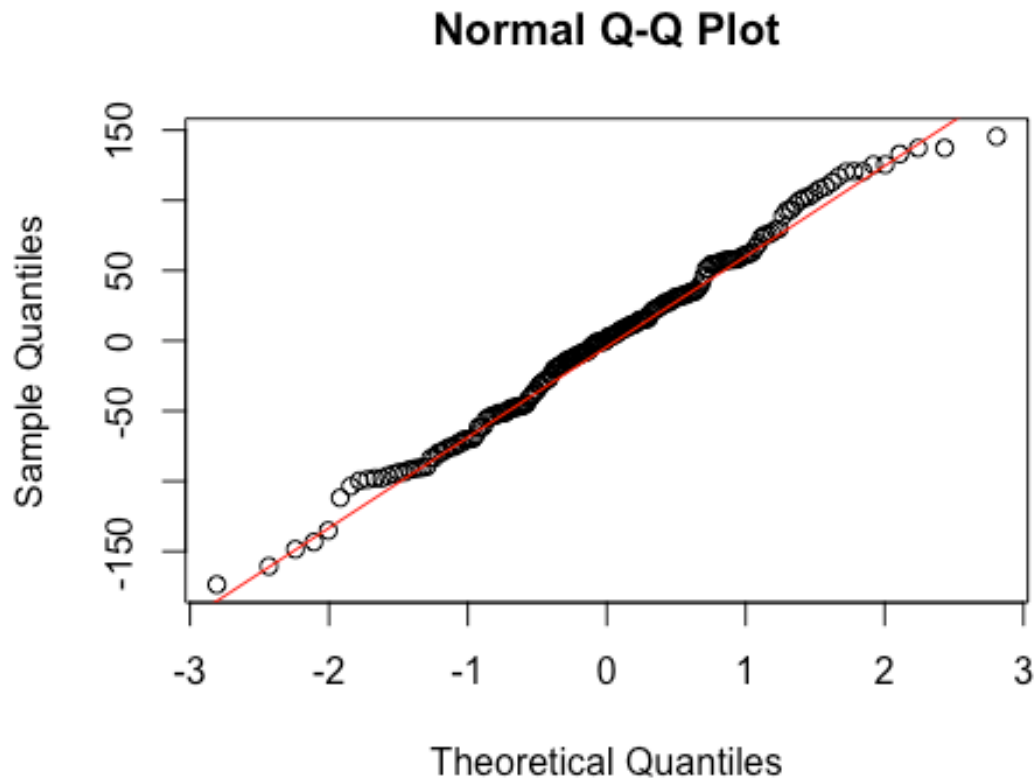


```r
#model the data
fit<-lm(Sales ~ Airplay, data = data)
summary(fit)


Call:
lm(formula = Sales ~ Airplay, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-173.649  -47.894    1.496   39.072  145.432 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  84.8725    11.2670   7.533 1.73e-12 ***
Airplay       3.9392     0.3743  10.524  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 64.79 on 198 degrees of freedom
Multiple R-squared:  0.3587,	Adjusted R-squared:  0.3555 
F-statistic: 110.7 on 1 and 198 DF,  p-value: < 2.2e-16

#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 35.87% of variance in album sales is explained by
airplay.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Airplay significantly and positively predicts album sales.
One unit increase in airplay is expected to be associated with an increase of
3.9392 units in album sales.

#interpretation: If airplay is zero, our album sales are expected to be
84.8725 (the intercept).

#If an album is played on the radio for 40 times, would be the predicted
album sales?
predict(fit, data.frame(Airplay = 40))

       1 
242.4398 
```

```r
#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```




```r
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```

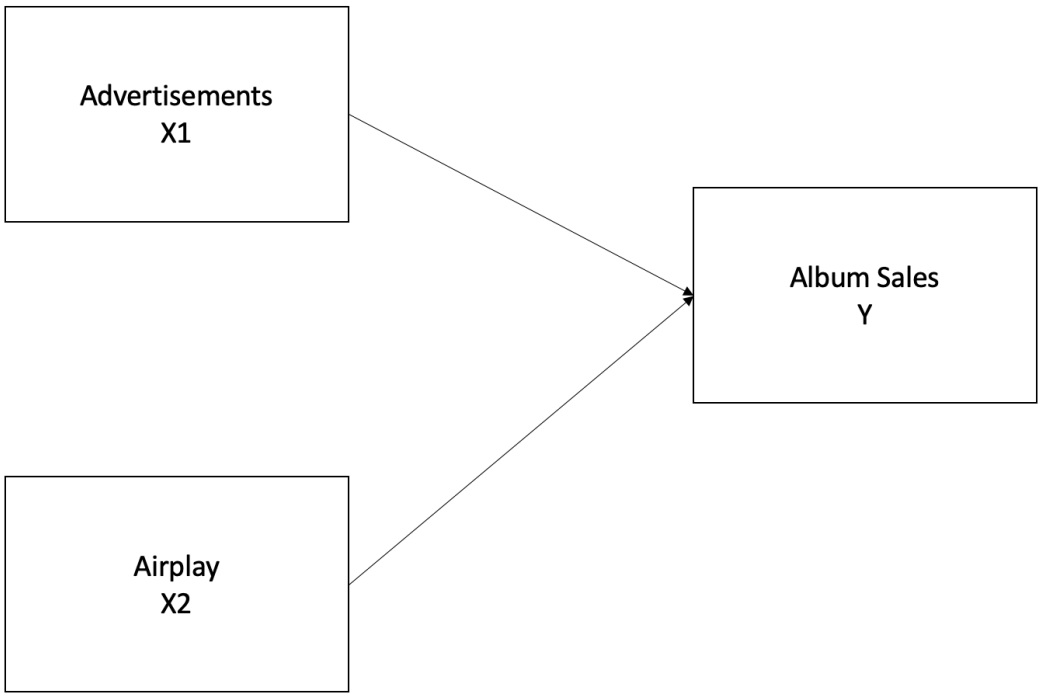


Multiple Linear Regression: > 1 predictor

Our world is complicated. A lot of things are determined by multiple factors. For instance, your health may be determined by how much water your drink per day, whether you smoke or not, how often you exercise per week, and how much sleep you have. It is quite natural that we want to find out the relative importance of each predictor to the dependent variable. This is when we use multiple linear regression by adding multiple predictors into the model.

Back to our album sales example. Suppose your company has been attempting to increase album sales through adverts and airplay for the past few years. Your boss wants to find out which channel has been more effective in increasing album sales. How can you find out the answer?

So, this is your model:



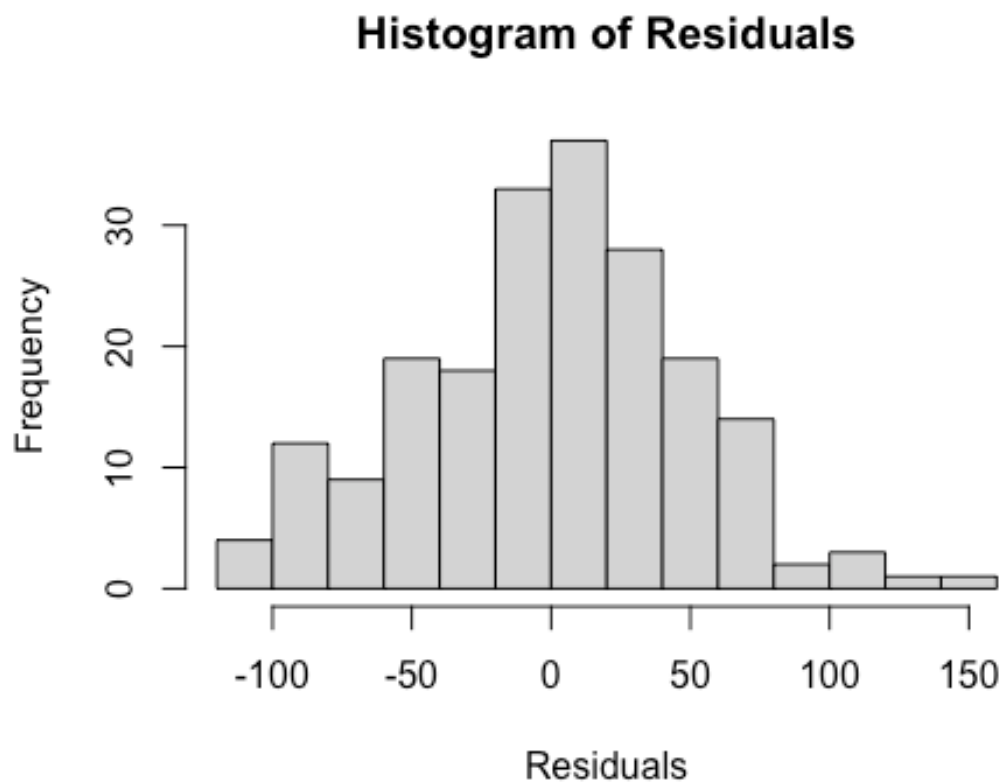
So, this is the equation:

$$Sales_j = i + \beta 1 Adverts_j + \beta 2 Airplay_j + \varepsilon_j$$

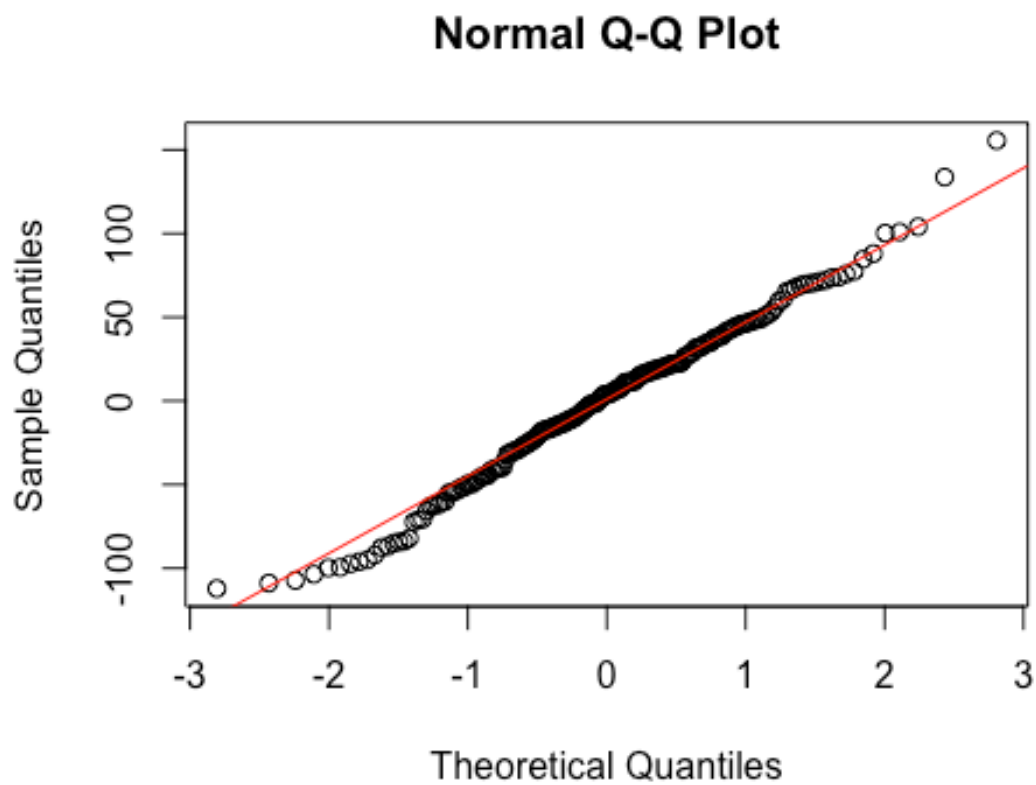
In this case, we care about both predictors (i.e., Adverts & Airplay). Therefore, Adverts and Airplay are considered **focal predictors** in our model. Now, let's use R to run the model and see what we find.

```
fit<-lm(Sales ~ Adverts + Airplay, data = data)
summary(fit)


Call:
lm(formula = Sales ~ Adverts + Airplay, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-112.121  -30.027    3.952   32.072  155.498 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 41.123811   9.330952   4.407 1.72e-05 ***
Adverts      0.086887   0.007246  11.991  < 2e-16 ***
Airplay      3.588789   0.286807  12.513  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 49.38 on 197 degrees of freedom
Multiple R-squared:  0.6293,    Adjusted R-squared:  0.6255
F-statistic: 167.2 on 2 and 197 DF,  p-value: < 2.2e-16

#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 62.93% of variance in album sales is explained by
adverts AND airplay.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Adverts significantly and positively predicts album sales.
One unit increase in adverts is expected to be associated with an increase of
0.086887 units in album sales.

#interpretation: Also, Airplay significantly and positively predicts album
sales. One unit increase in Airplay is expected to be associated with an
increase of 3.588789 units in album sales.

#interpretation: Suppose both predictors are measured on the same scale.
Airplay is more strongly related to album sales than Adverts.

#interpretation: If adverts and airplay are equal to zero, our album sales
are expected to be 41.123811 (the intercept).

#implication: Therefore, your boss probably wants to spend more money on
airplay than adverts.

#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```




```
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



Multiple Linear Regression: > 1 predictor

There is another scenario in which you may want to have more than one predictor in your regression model. Suppose based on your observation, you believe that an individual's income should positively predict an individual's happiness. That is, the more money you have, the happier you are. So, this is your model:

$$Happiness_j = i + \beta 1 Income_j + \varepsilon_j$$

However, previous studies have shown that health is strongly related to your happiness. Suppose we have two people, A and B. A is happier than B. A does have more money than B, but at the same time, A is healthier than B. In this case, how confident can you say income is related to happiness? It is very likely that the reason A is happier than B is his better health, not income. How can we solve this problem? In an ideal world, you may want to find a group of people with identical levels of health, but they differ in their income levels. Well, in a real world, it could be really difficult to find this group of people. Actually, you can achieve this goal simply by adding health into your regression model:

$$ Happiness_{j} = i + \beta1Income_{j} + {\color{red}\beta2Health_{j}} + \varepsilon_{j} $$

This model means: **holding health to be equal, does income predict happiness?** In this model, we only care about the relation between income and happiness. The decision to

bring health into the model is simply to control for its influence on happiness. Therefore, health is considered a **"covariate"** in this model.

Summary of Multiple Linear Regression

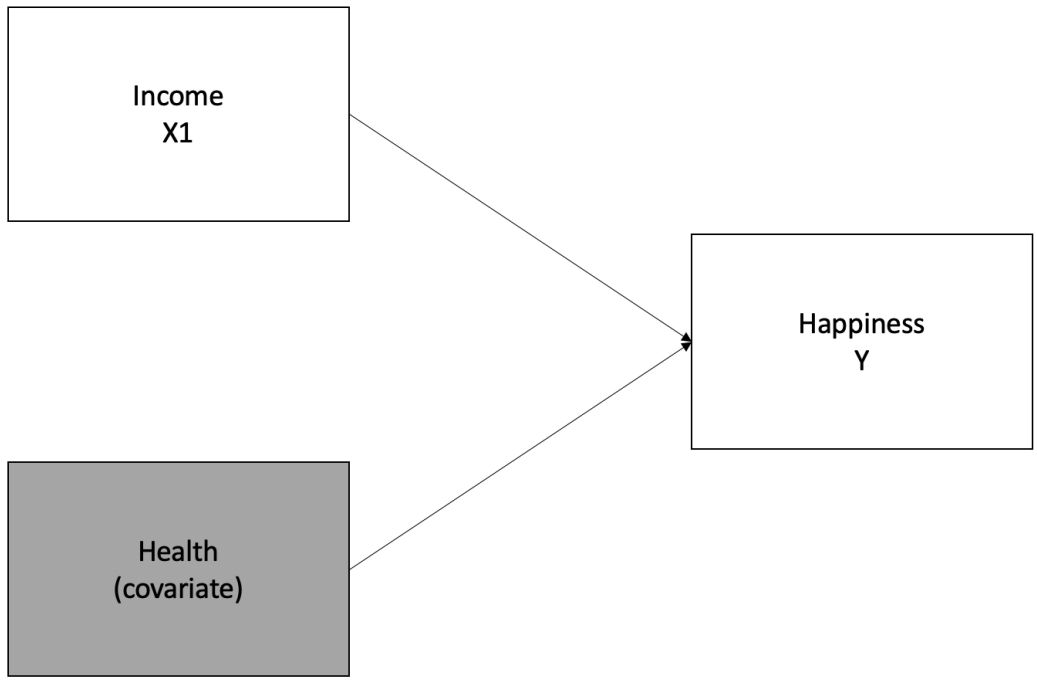
- There are two situations you would like to have more than one predictor in your regression model.
- Situation 1: When you want to examine the relative importance of a set of predictors. In this case, you care about all the predictors. Hence, they are called "focal predictors".
- Situation 2: When you want examine the relation between your focal predictor and the dependent variable but, at the same time, want to control for the influences of other predictors on your dependent variables. In this case, you don't care much about these "other predictors." Hence, they are called "covariates."

Activity

Let's examine the relation between income and happiness.

- Import the "happy" dataset.
- Draw the diagram to represent the relation.
- Visualize the relation between income and happiness. Is it positive or negative?
- Run a multiple linear regression model with income as the focal predictor, **health as the covariate**, and happiness as the dependent variable
- Find out the intercept and the slope.
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results.
- If you earn $50,000 per month with health index being 50, what would be the predicted happiness?
- If you earn $50,001 per month with health index being 50, what would be the predicted happiness?

Solution



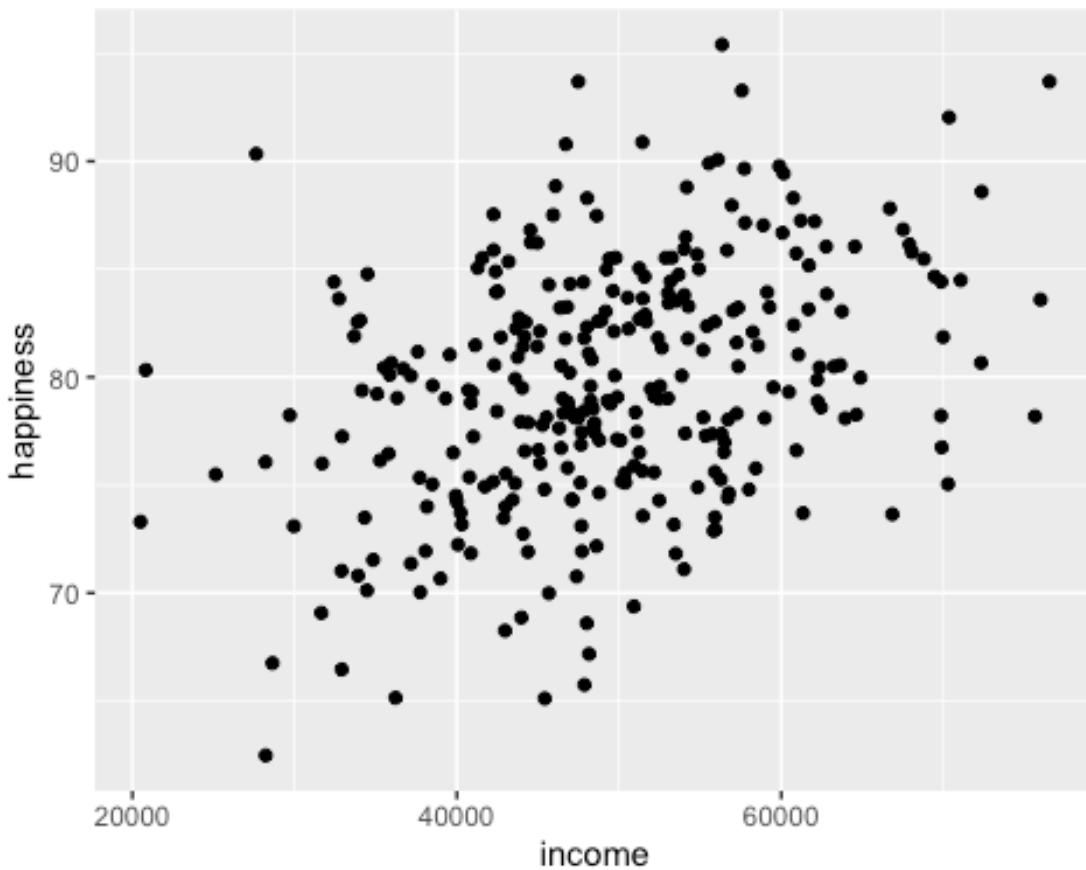

```
#import the dataset
happy <- read_sav("happy.sav")

#check our data
happy
```

```
# A tibble: 300 × 3
   income health happiness
    <dbl>  <dbl>     <dbl>
 1 48497.   77.0      77.9
 2 46722.  102.       90.8
 3 35518.   88.7      80.4
 4 43027.   77.7      75.5
 5 75985.   61.0      83.6
 6 49626.   77.1      84.0
 7 59135.   89.9      83.9
 8 48155.   83.7      67.2
 9 56098.   82.7      90.1
10 49473.   90.5      78.8
# ℹ 290 more rows
```

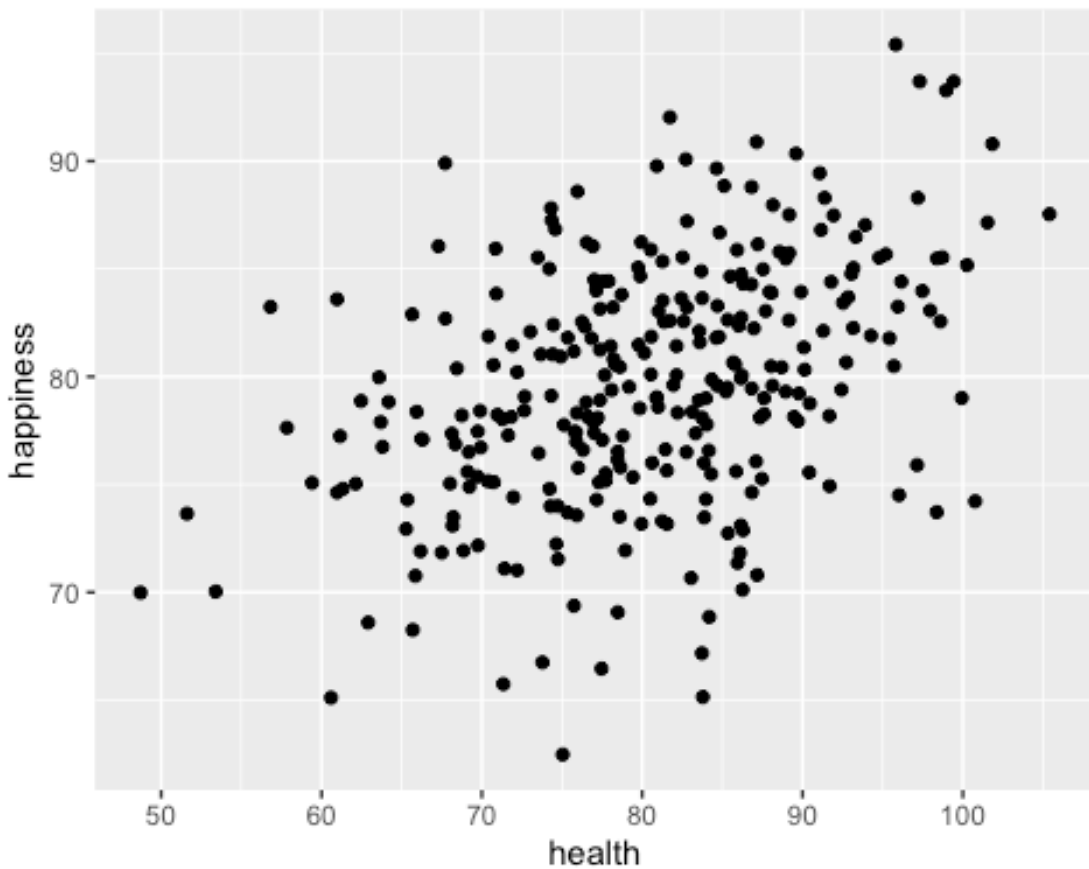
```
#descriptive statistics
describe(happy)
```

```
          vars   n     mean       sd   median  trimmed     mad      min
income       1 300 49435.28 10133.34 48656.49 49336.66 9436.45 20501.28
health       2 300    80.41     9.93    80.90    80.55    9.33    48.71
happiness    3 300    79.67     5.64    79.51    79.72    5.84    62.47
               max    range  skew kurtosis     se
income    76537.41 56036.13  0.08     0.05 585.05
health      105.40    56.69 -0.20    -0.03   0.57
happiness    95.41    32.93 -0.09     0.00   0.33
```

```
#visualize the data
happy |>
  ggplot(mapping = aes(x = income, y = happiness)) +
  geom_point()
```

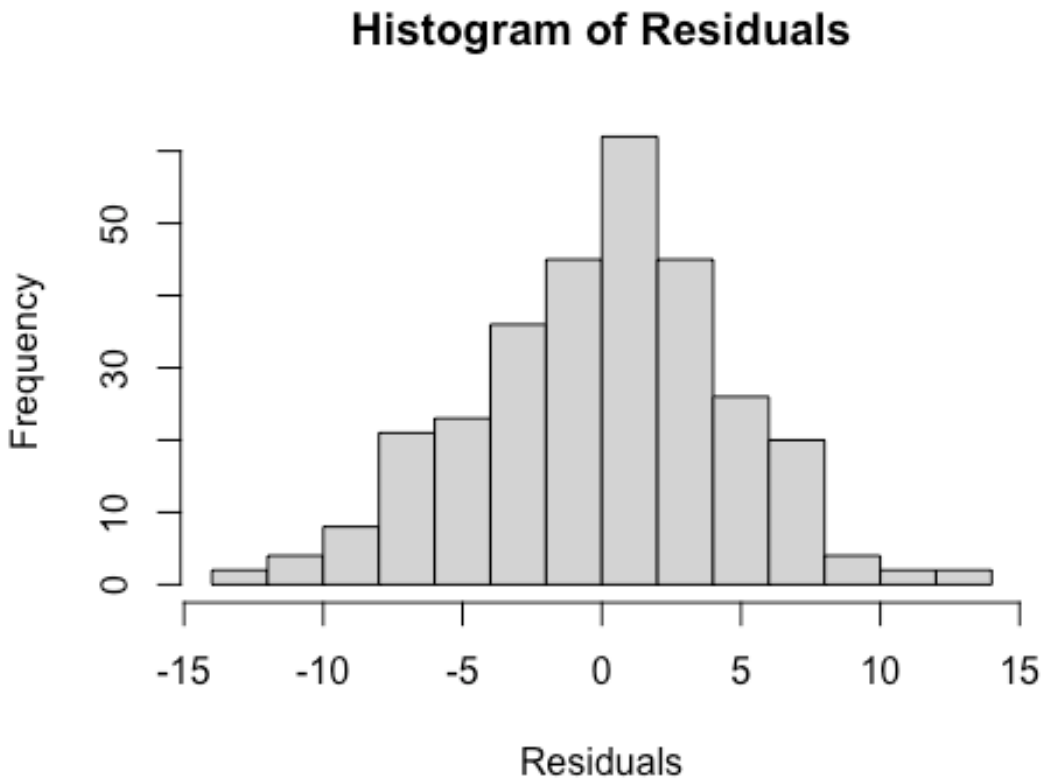



```
happy |>
  ggplot(mapping = aes(x = health, y = happiness)) +
  geom_point()
```

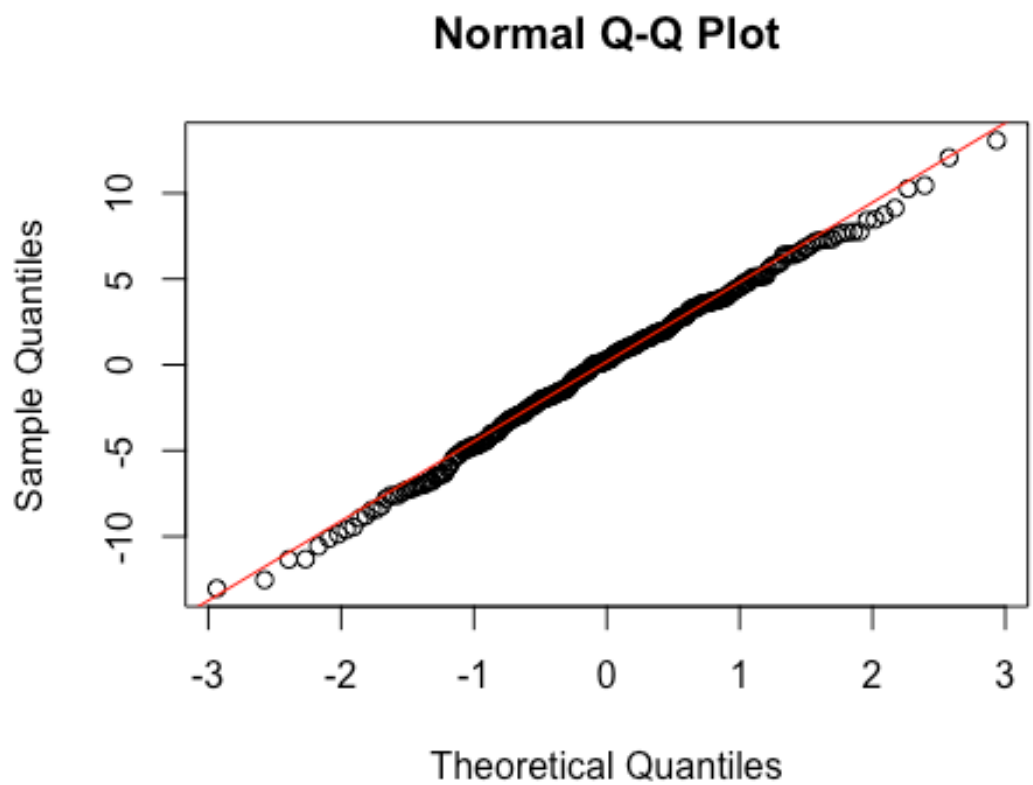



```
#model the data
fit<-lm(happiness ~ income + health, data = happy)
summary(fit)


Call:
lm(formula = happiness ~ income + health, data = happy)

Residuals:
     Min       1Q   Median       3Q      Max 
-13.0397  -2.9523   0.2712   3.3051  13.0644 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.920e+01  2.619e+00  18.786  < 2e-16 ***
income      2.138e-04  2.663e-05   8.027 2.34e-14 ***
health      2.475e-01  2.718e-02   9.103  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.661 on 297 degrees of freedom
Multiple R-squared:  0.3214,	Adjusted R-squared:  0.3168 
F-statistic: 70.34 on 2 and 297 DF,  p-value: < 2.2e-16
```

```
#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 32.14% of variance in happiness is explained by income
AND health.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Holding health to be equal, income significantly and
positively predicts happiness. One unit increase in income is expected to be
associated with an increase of 0.0002138 units in happiness.

#interpretation: If income and health are zero, your happiness is expected to
be 49.20 (the intercept).

#If you earn $50,000 per month with a health index being 50, what would be
the predicted happiness?
predict(fit, data.frame(income = 50000, health = 50))

      1
72.2597

#If you earn $50,001 per month with a health index being 50, what would be
the predicted happiness?
predict(fit, data.frame(income = 50001, health = 50))

       1
72.25991

#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```



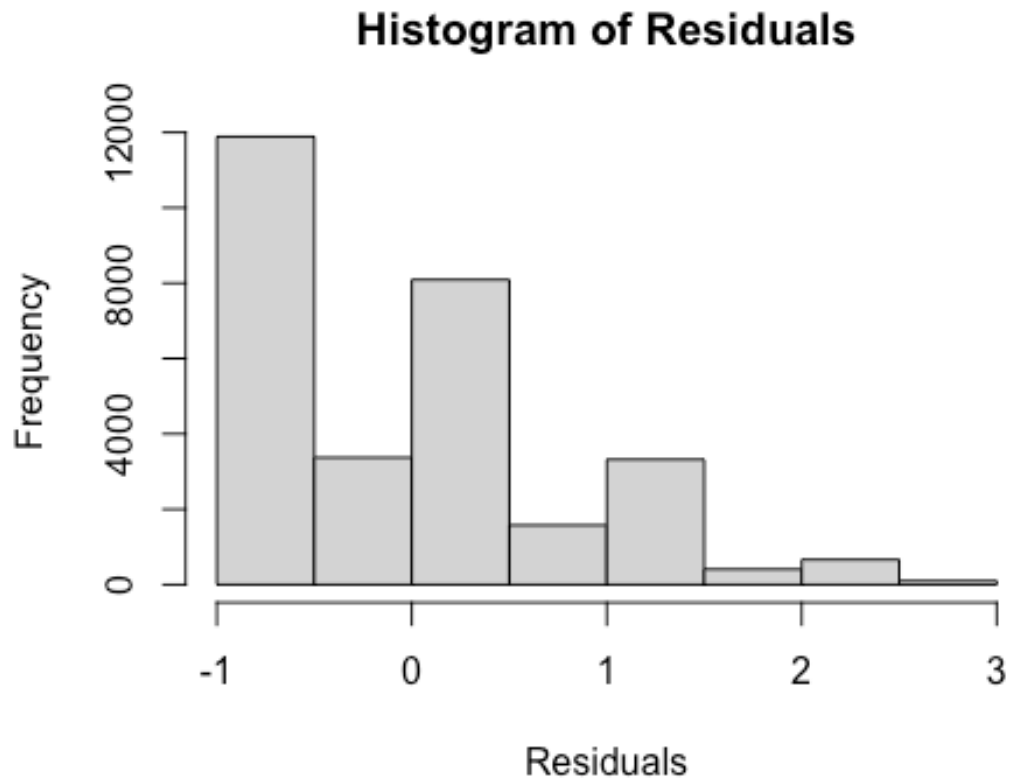

```
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



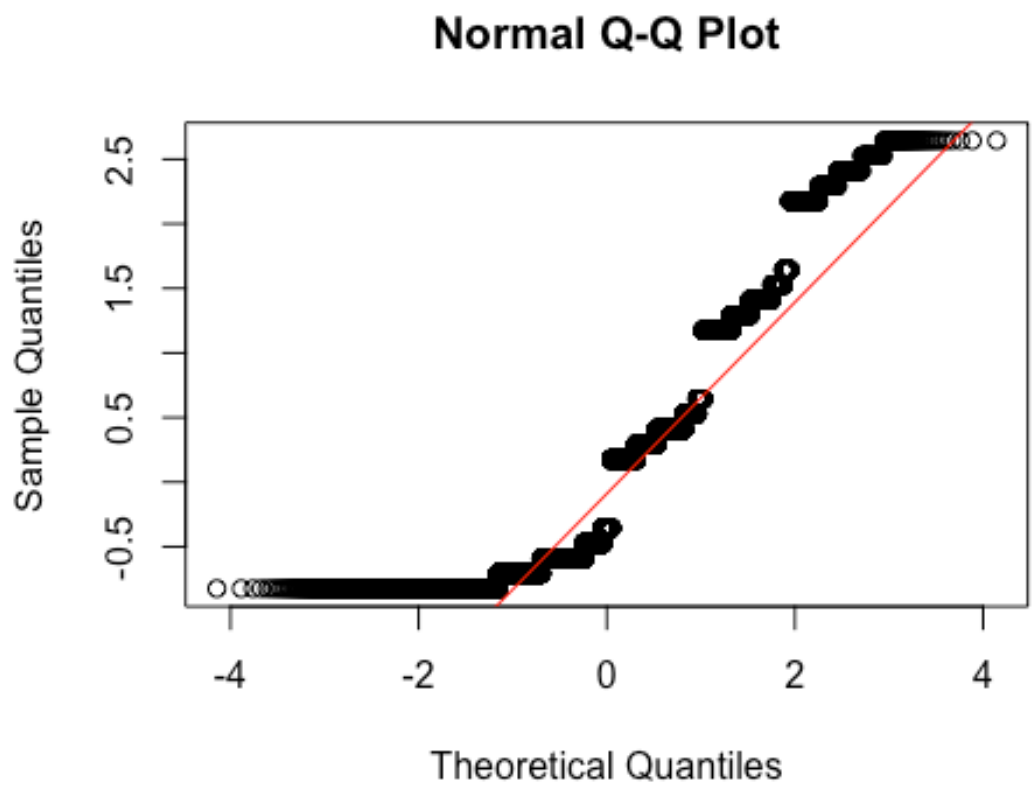
Bonus: How to Deal with Categorical Predictors in Multiple Linear Regression

```
#install.packages("fastDummies")
library(fastDummies)

Thank you for using fastDummies!

To acknowledge our work, please cite the package:

Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy
(Binary) Columns and Rows from Categorical Variables. Version 1.7.1. URL:
https://github.com/jacobkap/fastDummies,
https://jacobkap.github.io/fastDummies/.

# import data
data<-read_sav("nsch.sav")

# regression: X as a continuous predictor
#x = ScreenTime_21 (Time spent in front of a TV, computer, cellphone or other
electronic device watching programs, playing games, accessing the internet or
using social media, not including schoolwork )
#y = K7Q82_R (How often does this child (not) care about doing well in
```

```
school?)

table(data$ScreenTime_21)


    1     2     3     4     5 
 7625  8779 14047  9129 10353 

table(data$K7Q82_R)


    1     2     3     4    90    99 
15362  9742  3758   768 20719   543 

data<-data %>% filter(K7Q82_R < 5)

fit<-lm(K7Q82_R ~ ScreenTime_21, data=data)
summary(fit)


Call:
lm(formula = K7Q82_R ~ ScreenTime_21, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.8245 -0.5902 -0.3559  0.4098  2.6441 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   1.238731   0.014425   85.88   <2e-16 ***
ScreenTime_21 0.117151   0.003809   30.75   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7831 on 29412 degrees of freedom
  (216 observations deleted due to missingness)
Multiple R-squared:  0.03116,	Adjusted R-squared:  0.03112 
F-statistic: 945.9 on 1 and 29412 DF,  p-value: < 2.2e-16

#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```




```
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



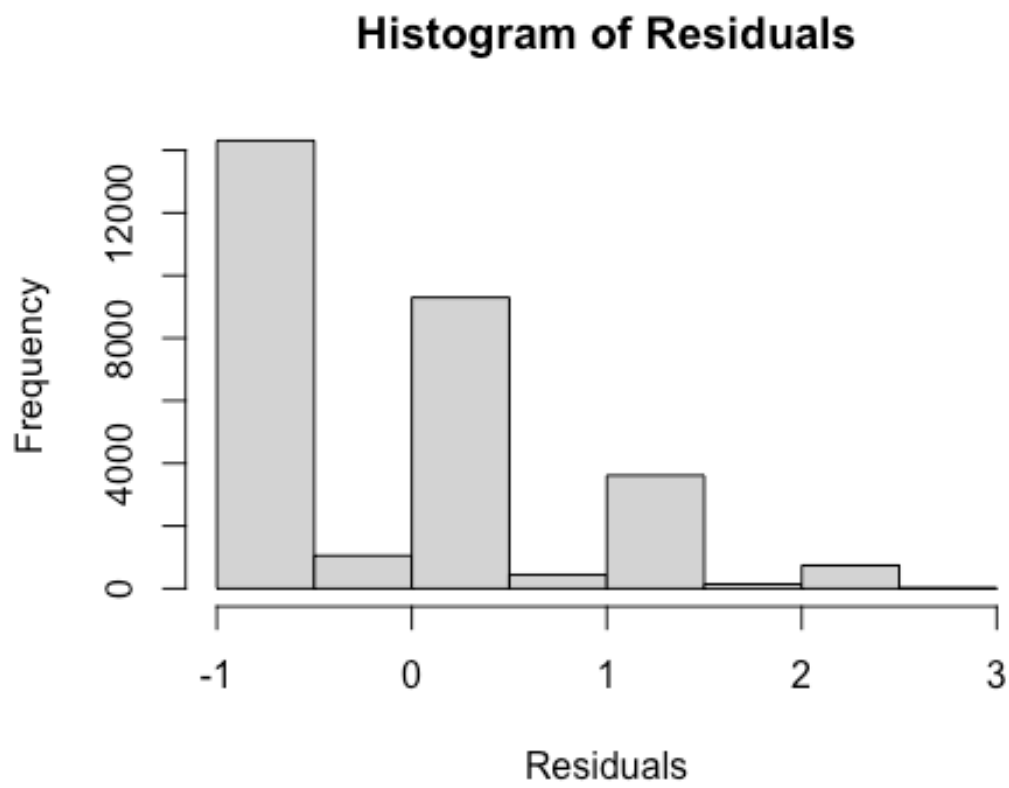

```
# regression: X as a categorical predictor
#x = race7_21 (racial group)
#y = K7Q82_R (How often does this child (not) care about doing well in
school?)

table(data$race7_21)


    1     2     3     4     5     6     7
 4054 19480  2021  1667   209    94  2105

#this is wrong
fit<-lm(K7Q82_R ~ race7_21, data=data)
summary(fit)


Call:
lm(formula = K7Q82_R ~ race7_21, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.6685 -0.6594 -0.6576  0.3406  2.3424
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.655775   0.008985 184.291   <2e-16 ***
race7_21    0.001822   0.003166   0.576    0.565
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.796 on 29628 degrees of freedom
Multiple R-squared:  1.118e-05, Adjusted R-squared:  -2.257e-05
F-statistic: 0.3313 on 1 and 29628 DF,  p-value: 0.5649
```

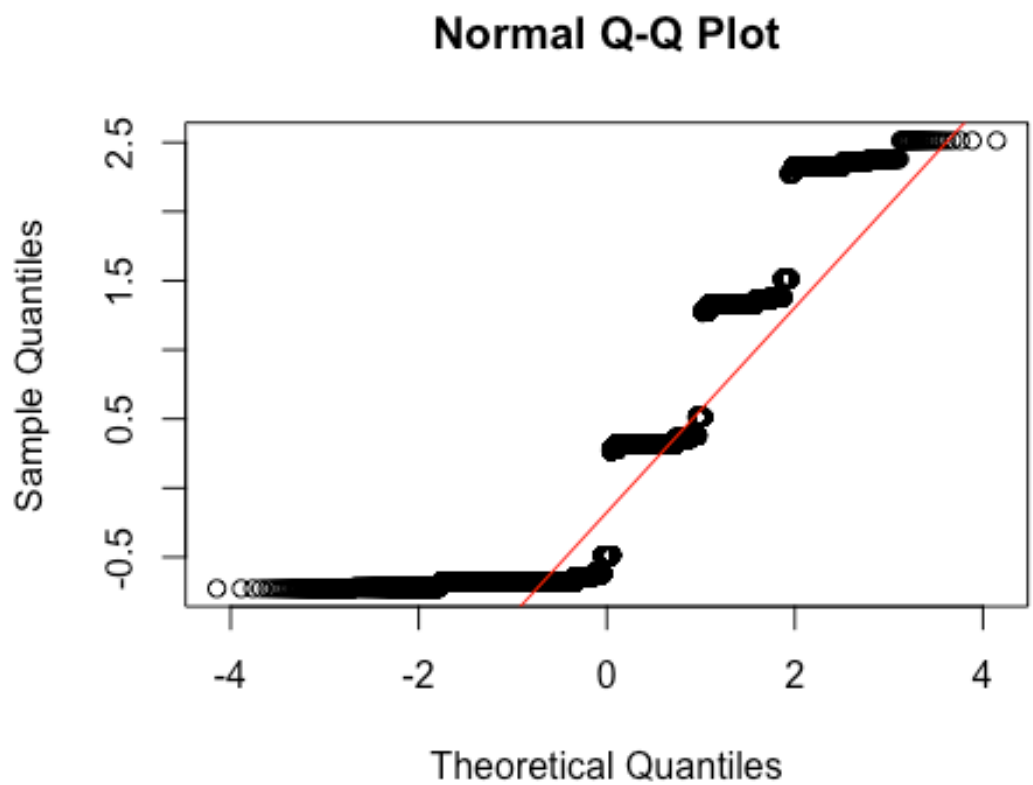
```r
#this is correct
data<-data %>% dummy_cols(select_columns = "race7_21")

fit<-lm(K7Q82_R ~ race7_21_1 + race7_21_2 + race7_21_4 + race7_21_5 +
race7_21_6 + race7_21_7, data=data)
summary(fit)
```

```
Call:
lm(formula = K7Q82_R ~ race7_21_1 + race7_21_2 + race7_21_4 +
    race7_21_5 + race7_21_6 + race7_21_7, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.7273 -0.6763 -0.4865  0.3237  2.5135

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.61999    0.01767  91.657  < 2e-16 ***
race7_21_1   0.01889    0.02164   0.873  0.38275
race7_21_2   0.05629    0.01857   3.032  0.00243 **
race7_21_4  -0.13349    0.02629  -5.078 3.84e-07 ***
race7_21_5   0.10728    0.05773   1.858  0.06314 .
race7_21_6   0.08214    0.08384   0.980  0.32723
race7_21_7   0.10020    0.02474   4.049 5.15e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7946 on 29623 degrees of freedom
Multiple R-squared:  0.003683,  Adjusted R-squared:  0.003481
F-statistic: 18.25 on 6 and 29623 DF,  p-value: < 2.2e-16
```

```r
#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```



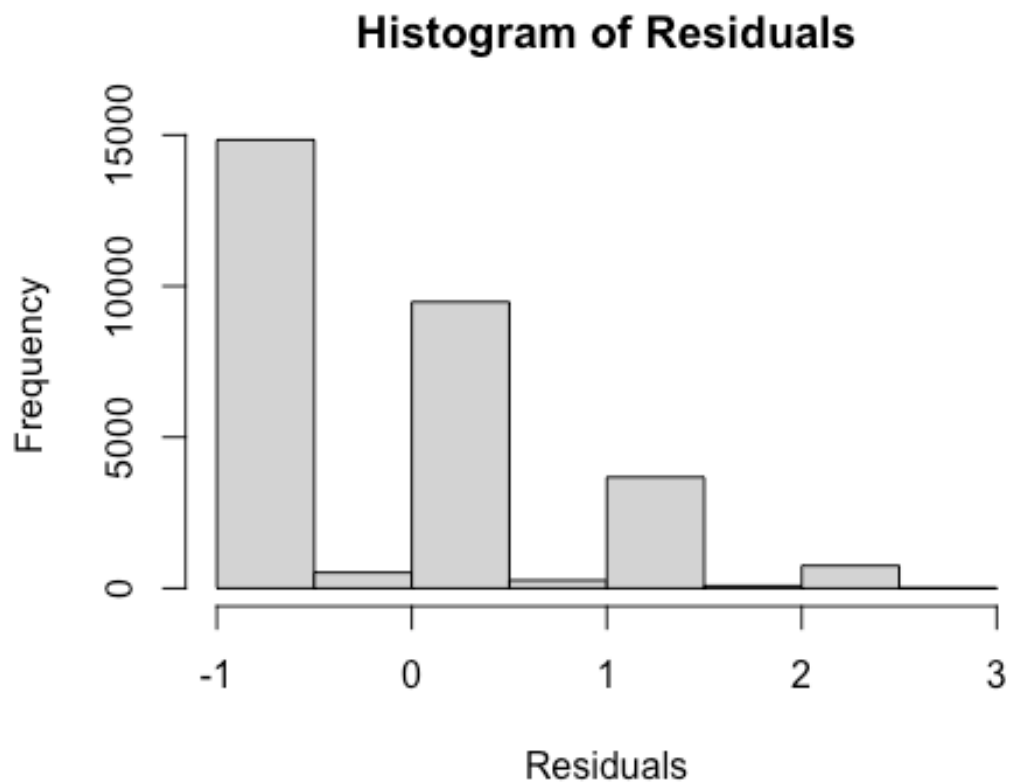

```
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



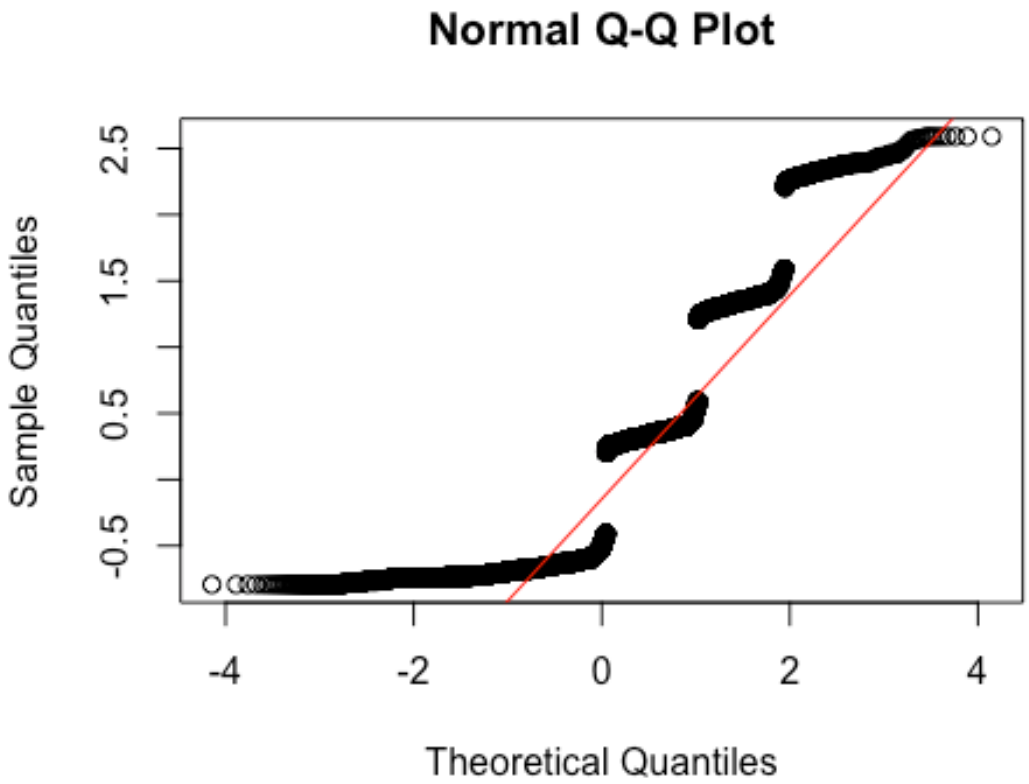

```
# regression: categorical predictor + continuous predictor

fit<-lm(K7Q82_R ~ race7_21_1 + race7_21_2 + race7_21_4 + race7_21_5 +
race7_21_6 + race7_21_7 + SC_AGE_YEARS, data=data)
summary(fit)


Call:
lm(formula = K7Q82_R ~ race7_21_1 + race7_21_2 + race7_21_4 +
    race7_21_5 + race7_21_6 + race7_21_7 + SC_AGE_YEARS, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.7932 -0.6662 -0.5019  0.3723  2.5898

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.463046   0.023546  62.136  < 2e-16 ***
race7_21_1    0.020622   0.021600   0.955  0.33972
race7_21_2    0.059047   0.018539   3.185  0.00145 **
race7_21_4   -0.131432   0.026245  -5.008 5.54e-07 ***
race7_21_5    0.106887   0.057635   1.855  0.06367 .
race7_21_6    0.081791   0.083695   0.977  0.32846
```

```
race7_21_7    0.107539   0.024714   4.351 1.36e-05 ***
SC_AGE_YEARS  0.013096   0.001301  10.066  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7932 on 29622 degrees of freedom
Multiple R-squared:  0.007079,  Adjusted R-squared:  0.006845
F-statistic: 30.17 on 7 and 29622 DF,  p-value: < 2.2e-16
```

```r
#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```



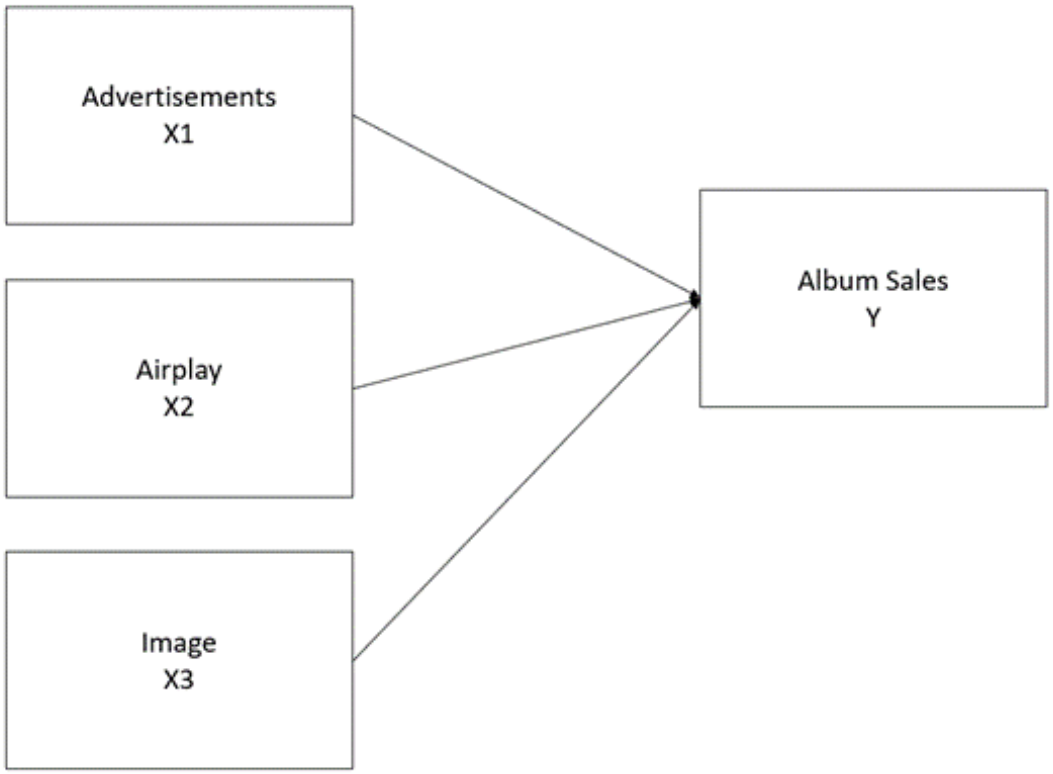

```r
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



Assignment 1

Use our album sales data. Suppose your company has been attempting to increase album sales through adverts, airplay, and image for the past few years. Your boss wants to find out which channel has been more effective in increasing album sales. Please run a multiple linear regression to find out the answer. (Let's assume that these three variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 1_LastFirstName."**

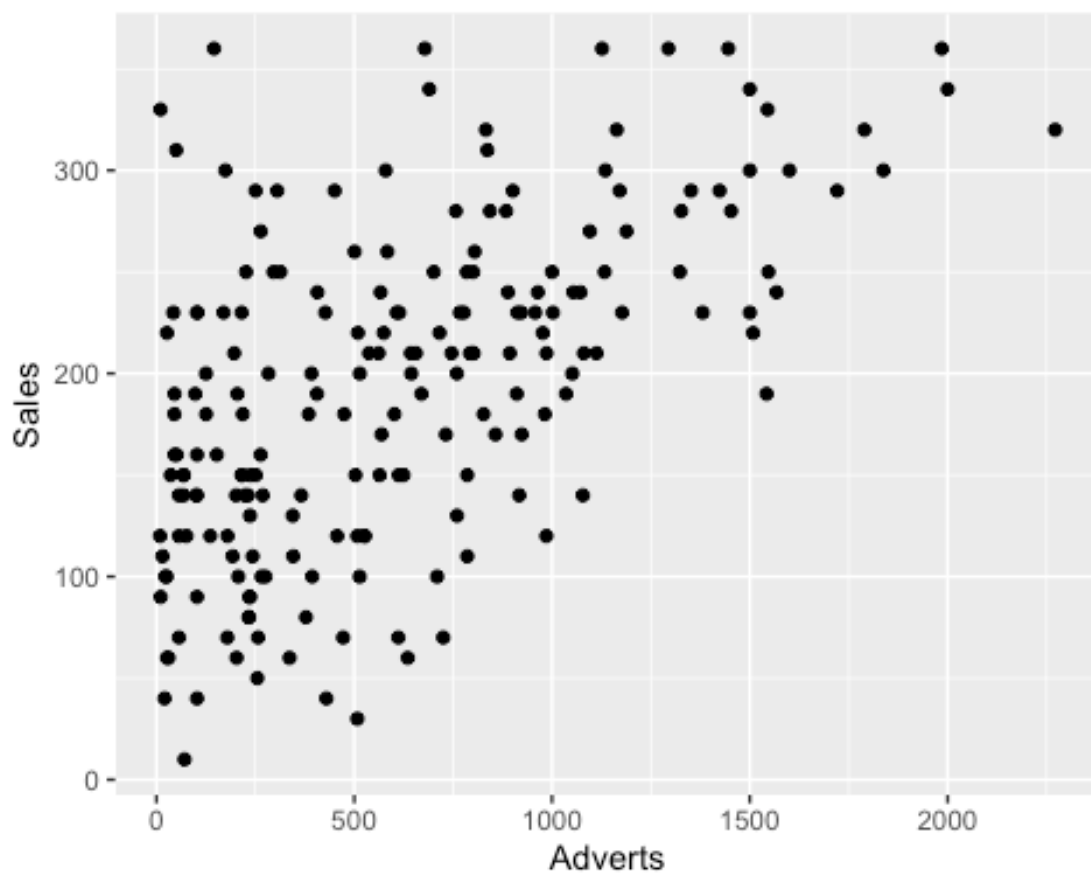
Solution



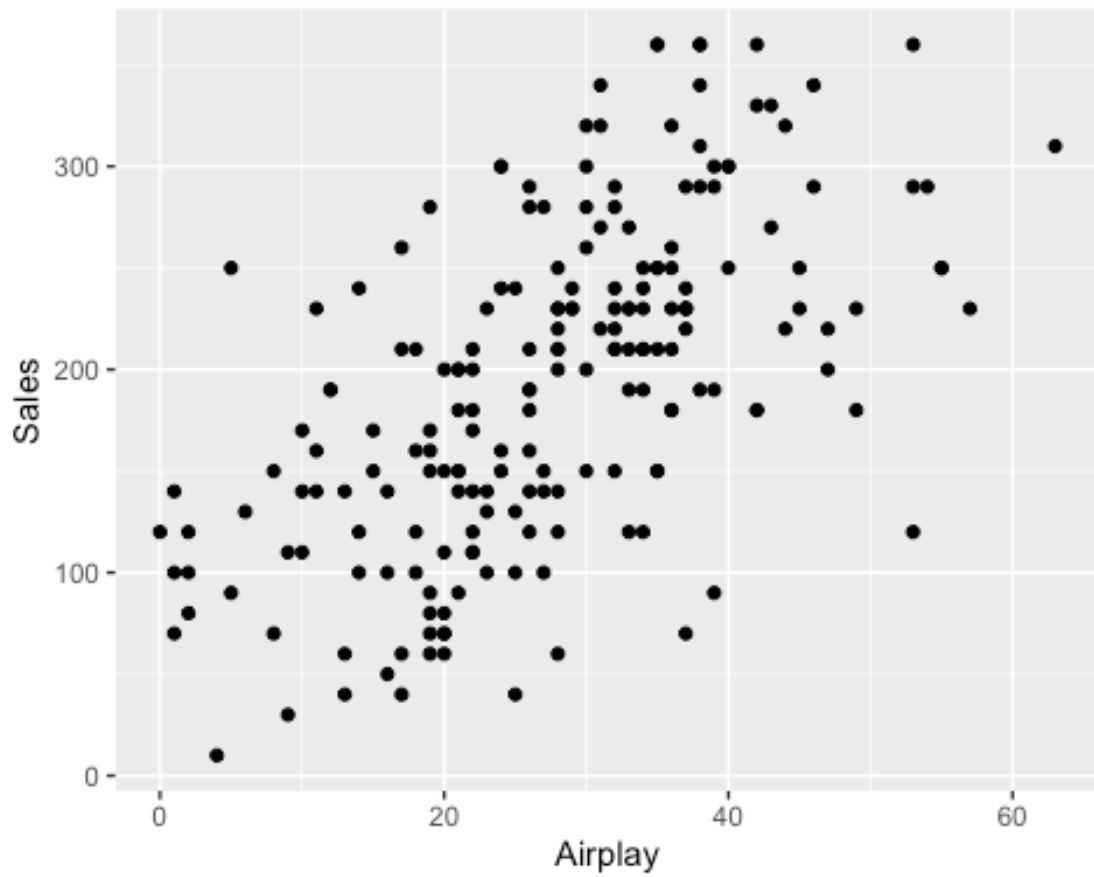

```
#import the dataset
data <- read_sav("Album Sales.sav")

#check our data
data

# A tibble: 200 × 4
   Adverts Sales Airplay Image
     <dbl> <dbl>   <dbl> <dbl>
 1    10.3   330      43    10
 2   986.    120      28     7
 3  1446.    360      35     7
 4  1188.    270      33     7
 5   575.    220      44     5
 6   569.    170      19     5
 7   472.     70      20     1
 8   537.    210      22     9
 9   514.    200      21     7
10   174.    300      40     7
# ℹ 190 more rows

#descriptive statistics
describe(data)
```

```
        vars   n   mean     sd median trimmed    mad  min     max   range
skew
Adverts    1 200 614.41 485.66 531.92  560.81 489.09  9.1 2271.86 2262.76
0.84
Sales      2 200 193.20  80.70 200.00  192.69  88.96 10.0  360.00  350.00
0.04
Airplay    3 200  27.50  12.27  28.00   27.46  11.86  0.0   63.00   63.00
0.06
Image      4 200   6.77   1.40   7.00    6.88   1.48  1.0   10.00    9.00 -
1.27
        kurtosis    se
Adverts     0.17 34.34
Sales      -0.72  5.71
Airplay    -0.09  0.87
Image       3.56  0.10
```

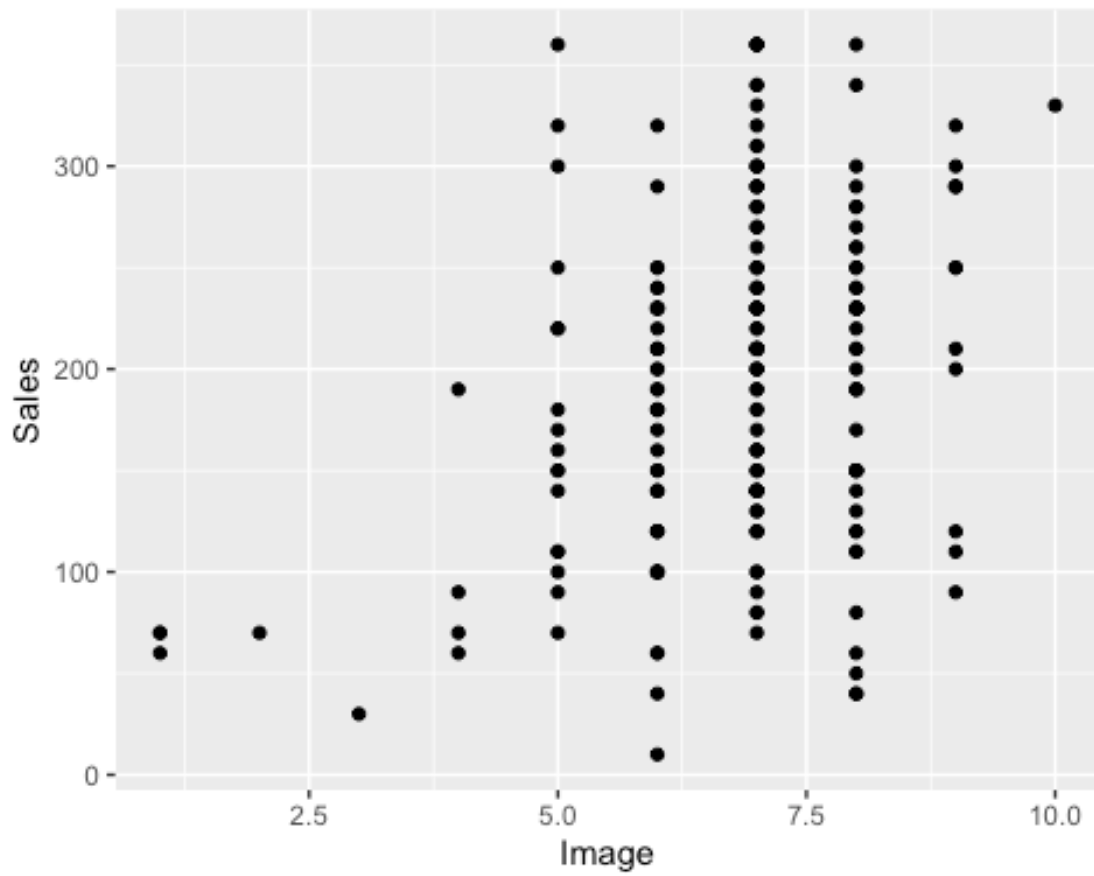
```
#visualize the data
data |>
  ggplot(mapping = aes(x = Adverts, y = Sales)) +
  geom_point()
```



```r
data |>
  ggplot(mapping = aes(x = Airplay, y = Sales)) +
  geom_point()
```

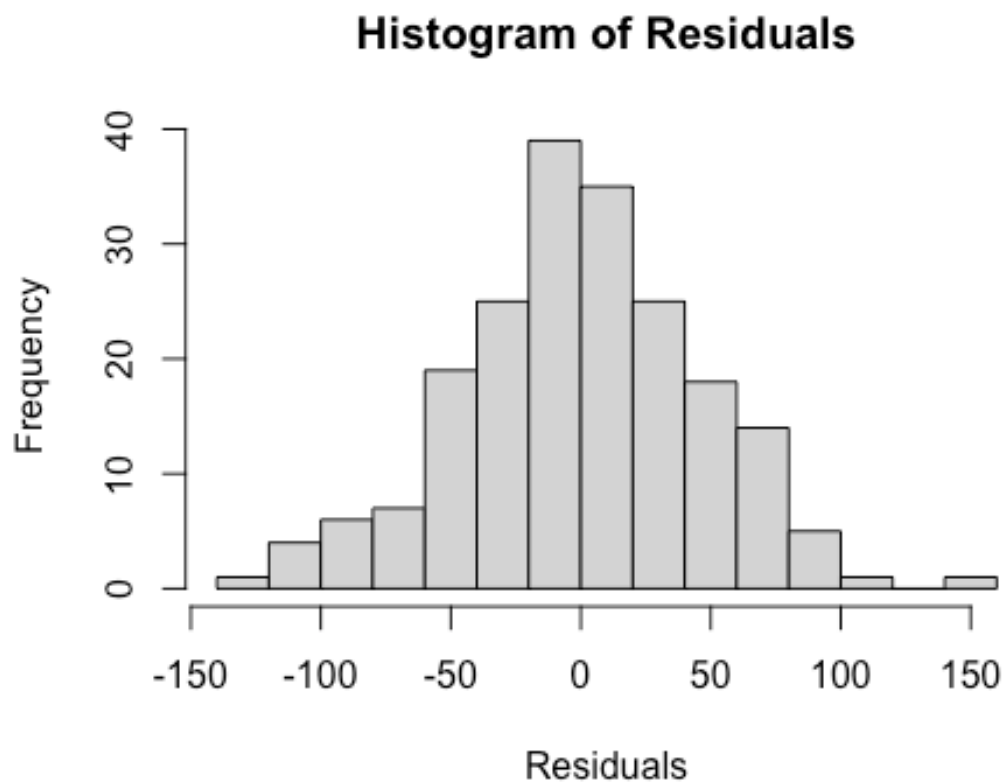



```r
data |>
  ggplot(mapping = aes(x = Image, y = Sales)) +
  geom_point()
```

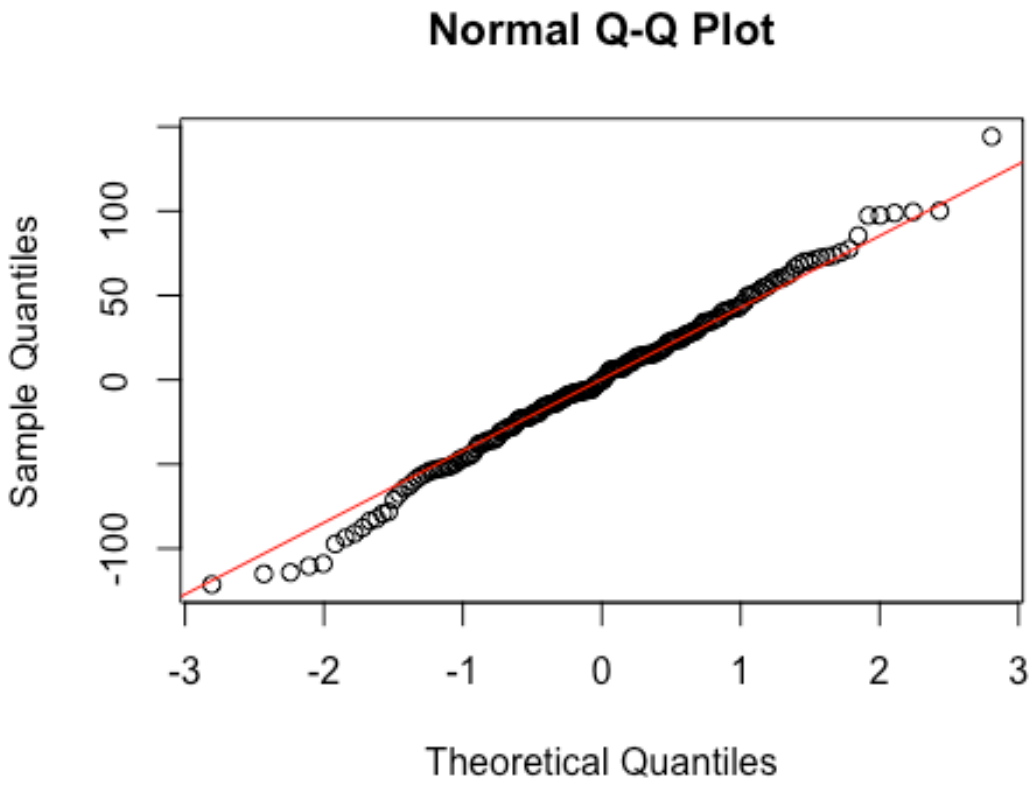



```
#model the data
fit<-lm(Sales ~ Adverts + Airplay + Image, data = data)
summary(fit)


Call:
lm(formula = Sales ~ Adverts + Airplay + Image, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-121.324  -28.336   -0.451   28.967  144.132 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -26.612958  17.350001  -1.534    0.127    
Adverts       0.084885   0.006923  12.261  < 2e-16 ***
Airplay       3.367425   0.277771  12.123  < 2e-16 ***
Image        11.086335   2.437849   4.548 9.49e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.09 on 196 degrees of freedom
```

```
Multiple R-squared:  0.6647,    Adjusted R-squared:  0.6595
F-statistic: 129.5 on 3 and 196 DF,  p-value: < 2.2e-16

#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 66.47% of variance in sales is explained by adverts,
airplay, and image.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Adverts significantly and positively predicts sales. One
unit increase in adverts is expected to be associated with an increase of
0.084885 units in sales after controlling for the effects of airplay and
image.
#interpretation: Airplay significantly and positively predicts sales. One
unit increase in airplay is expected to be associated with an increase of
3.367425 units in sales after controlling for the effects of adverts and
image.
#interpretation: Image significantly and positively predicts sales. One unit
increase in image is expected to be associated with an increase of 11.086335
units in sales after controlling for the effects of adverts and airplay.
#interpretation: Image is the most strong predictor of sales.
#interpretation: If adverts, airplay, and image are zero, sales is expected
to be -26.61 (the intercept).

#check residuals
# Plot histogram of residuals
hist(residuals(fit), main = "Histogram of Residuals", xlab = "Residuals",
breaks = 10)
```




```
# Q-Q plot
qqnorm(residuals(fit))
qqline(residuals(fit), col = "red")
```



Assignment: Problem and Theory Identification

- Identify a problem you would like to solve in your field.
- Identify a theory that can shed light on your problem and answer. Make sure that theory has a model.
- Introduce that model to your peers.
- Identify the part of the model you would like to empirically investigate.
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 3 mins.
- **Submit your PPT slides to the designated folder on iStudy.**